The paper appendices

Plan: Appendix A, the smooth model; Appendix B, the neural-network density model

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Abstract

The current draft appendix (`docs/paper/appendix_uncertainty.tex`, v4) describes the smooth model but still carries the old networks, ensembles and the Bayesian network as cross-checks, uses the word "discrepancy", and has figures with the old networks in them. We propose two self-contained appendices for the paper, describing only the two models to be published: **Appendix A**, the smooth spline model, which is the model of the paper; and **Appendix B**, the neural-network density model, as an independent model of the same quantity. Everything else (the bin-value networks, ensembles, the Bayesian network, the Laplace mixture) is left out. The plan fits a two-week schedule; decisions for you are in section 6.

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1 Principles

- **Two models only.** Each appendix describes one model completely: what it is, how it is fitted, its uncertainty, and how well it does. B refers to A for everything they share (the data, the measures, the model-error term) and does not repeat it.
- **One language.** The likelihood-ratio statistic $-2\ln\lambda$ for accuracy; pulls for calibration; "model-error term" for the systematic uncertainty, introduced in A as a systematic uncertainty in the physicists' sense. "Discrepancy" and "deviance" disappear.
- **Every number from a script in the repository.** Each figure and table is regenerated from the committed model files (`runs/SMOOTH_J40`, `runs/FTD_SiLU_huber_d2_pearson_curv1e-06`), with no number copied from an old workspace.

- **For the reader of the paper**: a particle physicist who knows HEFT and Poisson statistics but not splines or quadrature. Concise, with the pedagogy of the notes condensed into one sentence each where needed.

2 Appendix A: the smooth model

Target length: 5–6 journal pages. Sources: `dihiggs_model.pdf` (sections 3–8), the current appendix, `smooth_results.pdf`.

A.1 The form and the data. The 23-term form $d\sigma/dm = \sum_i C_i(\theta) a_i(m)$ and when it is exact (no θ dependence anywhere else in the calculation). The POWHEG spectra as exact Poisson counts ($t_{sk} = q_s n_{sk}$); the fit rows, the quick-test rows and the 1,000 test spectra.

A.2 The model. Cubic B-splines times the threshold factor, knots uniform in $\sqrt{m - 250}$; the prediction is the bin integral, linear in the 920 coefficients. The free fit as the limit with one box per bin, which sets the floor. *Figure:* `fig_basis`.

A.3 Fit and posterior. Poisson maximum likelihood by iterated weights from the model (why not the observed counts, in one sentence); the second-difference prior with strengths from the evidence; the exact Gaussian posterior; its cost.

A.4 The systematic uncertainty. The three uncertainties (Poisson noise, fit, model error); the floor reached by both the smooth model and the free fit; its growth with cancellation and its smoothness in m; why the errors must vary from run to run; the form $\varepsilon(m) \sum_i |C_i A_{ik}|$ as the largest spread relative errors ε can produce; ε from the quick-test rows (3 parameters); what it does not include (scales, PDFs, top-mass scheme). *Figure:* `fig_model_error` *(from* `dihiggs_model.pdf`*):* $\varepsilon(m)$*, pull width against* m*,* $\langle z^2 \rangle$ *by cancellation fifth for four treatments.*

A.5 Validation on the test spectra. Accuracy (table: smooth model and free fit; figure against m), calibration (table), three example spectra (`fig_spectra_dz`, replacing the nine-panel figure), inside a bin (merged-bin test), the edges (240–255 and 1455–1500 GeV).

A.6 The coefficient functions. The 23 $a_i(m)$ with their exact bands, *without* the old network's points (new figure).

A.7 Using the model. What is published (the model file, the code), the valid range 250–1455 GeV, the uncertainty for one bin and for sums of bins, and the SM example (28.11 fb in 255–1455 GeV; fit uncertainty 0.006 fb; model-error term 0.03–0.17 fb).

3 Appendix B: the neural-network density model

Target length: 3–4 journal pages. Sources: `density_plan.pdf` (with addendum), `density_results.pdf`.

B.1 Why a second model. An independent representation of the same 23 functions, built in an entirely different way, tests whether the results of A depend on the choice of splines and prior.

B.2 The network. $a_i(m) = \phi(m)\, P_i(z)\, e^{Q_i(z)}$: P a six-layer network with SiLU activations and a tanh output, Q linear, input $z = 2\sqrt{(m - 250)/1205} - 1$; 3,944 parameters.

B.3 Bin integrals by Gauss–Legendre quadrature. Eight nodes per bin; why the quadrature must be far more accurate than the noise (its error becomes a bias in the density); the cost (640 evaluations per step, independent of the batch). One paragraph on why a network trained on bin contents at bin centres is not a density (biased by about 1% near threshold on a realistic toy), without describing the old networks.

B.4 Training. L1 loss, then the Huber loss of the pulls with Poisson variances from the model; the smoothness penalty (curvature of $P_i e^{Q_i}$ in z, $\lambda = 10^{-6}$) and why it is needed (the poorly determined a_{16}–a_{19}).

B.5 Results. Accuracy (table: A, free fit, B); calibration with its own ε, which equals A's: the systematic is a property of POWHEG, not of either model; agreement of the densities (0.31% median, equal at and between bin centres); the coefficient functions against A's bands. *Figures: accuracy against m for A and B; $\varepsilon(m)$ of both; $d\sigma/dm$ of A and B for three spectra with the ratio B/A below (new, without the old networks); the 23 a_i of B over A's bands.*

B.6 Limitations. A single network (no fit uncertainty of its own, unless the seeds ensemble is done); slightly less accurate than A just above threshold.

4 Files

- `docs/paper/appendix_A_smooth.tex`, `docs/paper/appendix_B_density.tex`: the two appendices, each to be `\input` after `\appendix` in the paper (JHEP style), with `appendix_refs.bib`.
- `docs/paper/appendices_draft.tex`: a stand-alone wrapper that compiles both, for reading.
- Figure scripts in `docs/paper/`, one per figure, all reading the repository's model files through `smooth_common.py`. New: `fig_A_coefficients.py`, `fig_A_merged.py` (ported from `docs/smooth/scripts/merged.py`), `fig_B_accuracy.py`, `fig_B_density.py`, `fig_B_coefficients.py`, `fig_B_epsilon.py`. Reused: `fig_basis.py`, `fig_spectra_dz.py`, `docs/model/model_error_study.py`.
- The current appendix (v4) moves to `docs/paper/v4_smooth/`, like v2 and v3; the executive summary is updated at the end.

5 Schedule (two weeks)

days 1–3	Appendix A: text, new figures, the merged-bin script ported; you review
days 4–5	Appendix B: text and figures; you review
in parallel (5090)	optional: seeds ensemble and merged-bin test for B (section 6)
days 6–8	revisions from you and the colleagues; references; consistency with the main text
days 9–10	executive summary; final numbers; tagged commit

6 Decisions for you

Decisions before I proceed

1. **Structure**: A and B as in sections 2 and 3. Agreed?
2. **The old networks**: left out entirely, except one sentence in B.3 that a network trained on bin contents at bin centres does not give the density (with the toy result). Agreed, or leave even that out?
3. **Seeds ensemble for B** (10 seeds of L1 + fine-tune on the 5090, a few hours): gives B a fit uncertainty and a measured spread between bin centres. Worth it for the paper, or state the limitation?
4. **Merged-bin test for B** (one L1 + fine-tune on 30 GeV bins): makes B's claim to be a density as strong as A's. Worth it?
5. **The main text**: who writes it, and which section labels should the appendices refer to (now `sec:model`, a placeholder)? Should the appendices assume the main text has already introduced the 23-term form and the POWHEG samples, or repeat them briefly?
6. **Document class**: JHEP (`jheppub`) for the wrapper too, or a plain article for drafts?